

ECON 21030: Honors Econometrics

Homework 2

Spring Quarter 2024

Question 1: [10 points] Consider an outcome Y and a discrete random variable X with K points of support, so that $X \in \{1, \dots, K\}$. Define $X_k = 1\{X = k\}$.

1. Derive an explicit expression for β_k in the regression $Y = \beta_1 X_1 + \dots + \beta_K X_K + U$ using the usual matrix-algebra definition of regression coefficients. Interpret it.
2. Could I have included a constant in this regression as well? Why or why not?
3. Suppose I include a constant but omit X_1 . Derive an expression for β_k for $k > 1$ in this case. You may find it helpful to start with the $K = 2$ case.
4. How would β_k be affected if we omitted $X_j, j \neq k, j > 1$ instead?
5. Is $\hat{Y} = X'\beta$ affected by which indicator X_j you choose to drop?
6. Now suppose we have an iid $\{(Y_i, X_i)\}_{i=1}^N$? What form does the OLS estimator of β take in the case where I transform X_i into k indicator variables and omit a constant? What about when we drop the j th indicator and include a constant?
7. Derive the sampling variance of β_k in the case with no constant and without assuming homoskedasticity. Consider the plug-in estimator (i.e., EHW). Is it unbiased? How does it relate the variance of a sample mean?
8. What about the HC2 estimator? Can you show whether it is biased? Does it depend on homoskedasticity?

Question 2: [5 points] Prove that the OLS estimator $\hat{\beta}$ and sample residuals $\hat{U} = \mathbf{Y} - \mathbf{X}'\hat{\beta}$ are uncorrelated if $E[\mathbf{U}|\mathbf{X}] = 0$. Define correlation in this case as $E[\hat{\beta}\hat{\mathbf{U}}'] - E[\hat{\beta}]E[\hat{\mathbf{U}}']$. Why is this useful?

Question 3: [5 points] Prove the following version of the population Frisch-Waugh-Lovell theorem (note: there is a hard way to do this, involving matrix algebra, and an easy way involving properties of regression residuals):

1. The coefficient β in the regression $Y = X'\beta + W'\gamma + U$ is equivalently given by the coefficient in the regression $Y = \tilde{X}'\beta + V$, where $\tilde{X} = X - (W'E[WW']^{-1}E[WX'])'$. Assume W includes a constant.
2. β is also given by the regression $\tilde{Y} = \tilde{X}'\beta + V$, where $\tilde{Y} = Y - W'E[WW']^{-1}E[WY]$.

Question 4: [5 points] Consider a standard OLS estimator for an iid sample and the linear restriction $R\beta = r$.

1. Find a closed-form solution for the estimator of β that minimizes the sample sum of squares subject to the constraint that $R\hat{\beta} = r$.
2. Show that the F statistic:

$$F = \frac{(R\hat{\beta} - r)'[R(\mathbf{X}'\mathbf{X})^{-1}R']^{-1}(R\hat{\beta} - r)/m}{\hat{\sigma}^2}$$

can also be written as:

$$F = \frac{(SSR_R - SSR_U)/m}{SSR_U/(N - K)}$$

where SSR_U is the sum of squared residuals from the unrestricted model, SSR_R is the sum of squared residuals for the restricted solution $\tilde{\beta}$ that satisfies $\min_{\tilde{\beta}} SSR(\tilde{\beta})$ s.t. $R\tilde{\beta} = r$, and K is the dimension of β .

3. Derive an expression relating the R^2 of a regression to the F test of the hypothesis that all regressors besides the constant are equal to zero. Why is this useful?

Question 5: [15 points; warning, hard] Consider two random variables: $X \sim N(0, 1)$ and $Y = X^3$.

1. What is the conditional expectation of Y given X ?
2. What is the coefficient on X from the population regression of Y onto X and a constant? You may need to compute (or look up) the uncentered moments of a standard normal distribution.
3. What is the residual from the population projection? What is $E[XU]$? Is U independent of X ?
4. Suppose we have an iid sample (Y_i, X_i) of size N . We're going to investigate the bias and consistency of the OLS estimator of β . To start, use the covariance-variance representation of $\hat{\beta}$ to write down $\hat{\beta}$ as a function of sample averages of powers of X , i.e., $\mu_X = \frac{1}{N}X$, $\mu_{X^2} = \frac{1}{N}X^2$, etc. Is $\hat{\beta}$ consistent for β ?
5. Write a short simulation that computes $\hat{\beta}$ for $N \in \{10, 50, 100, 100\}$. What do you notice about bias?
6. For simplicity, assume that the sample average of X , μ_X , is exactly zero. You should see that in this case $\hat{\beta}$ is the ratio of two sample moments of X . Are each of those sample moments unbiased for their respective population moments?
7. Write down a second order Taylor expansion for a function of two random variables $f(X, Y)$ around $f(E[X], E[Y])$. Take its expectation and interpret the results.
8. Evaluate this expression using $f(X, Y) = X/Y$. Use the sample moment in the numerator of your expression for $\hat{\beta}$ as X and the sample moment in the denominator as Y .
9. Finally, compute any variance-covariance terms in this expression to derive a formula for the bias of $\hat{\beta}$. It should involve only a constant and N , and should come close what you found numerically above.
10. What have we learned about the bias of OLS here? What requirements for OLS to be unbiased for the population projection coefficient are we missing?